MTH 346/646 homework cover sheet

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What topics did this assignment cover? How comfortable do you feel with each of those topics?

It covered finite generation of curves, or the lack thereof. I feel quite comfortable with the theory.

In the course of doing this assignment, where did you get stuck? How did you get unstuck?

I wouldn't say I got particularly stuck at any point, though I did have to try a few different things to figure out how to force 2(b).

Is there anything you've done here that you're not confident about, that you want me to take a really close look at, or that you want me to suggest an alternate approach to?

I'm really not sure how to find the second generator in 1, or even how to prove that it can be generated by only 2 elements.

1. Let $E: y^2 = x^3 + x^2 - 2x + 1$. The goal of this problem is to use Magma to find two points P and Q so that $E(\mathbb{Q})$ is generated by P and Q. (You may use Magma for computations, but don't use the commands `Generators` or `MordellWeilGroup`).

The information below gives you explicit forms of the four lemmas.

(a) You may assume that $[E(\mathbb{Q}) : 2E(\mathbb{Q})] = 2$ and coset representatives are $Q_1 = (0:1:0)$ and $Q_2 = (-2, 1)$.

(b) You may assume that if $P_0 \in E(\mathbb{Q})$, then for all $P \in E(\mathbb{Q})$ we have $h(P_0 + P) \leq 2h(P) + 2h(P_0) + 3.8821$. Also, for all $P \in E(\mathbb{Q})$, $h(2P) \geq 4h(P) - 3.8821$.

(c) The Magma command `NaiveHeight` of a point P returns $h(P)$. Also, the command `Points(E : Bound := B)` returns all points P with $h(P) \leq \log(B)$.

Proof. We first want to find, in the language of the Descent Theorem, our κ' and κ. The statement of (b) above tells us immediately that $\kappa = 3.8821$. Also by (b), κ_i for a given $P_i \in E(\mathbb{Q})$ is $2h(P_i) + 3.8821$, so we want to take the max of the κ_i's corresponding to the Q_i's. With this notation, Magma tells us that $\kappa_1 = 3.8821$ and $\kappa_2 \approx 5.2684$. Thus we can define $\kappa' = 5.2684$. The number we most want out of this is $\kappa + \kappa' = 9.1505$.

We use this to conclude that if we have a $P \in E(\mathbb{Q})$ such that $h(P) \geq 9.1505$, then we can find an $R \in E(\mathbb{Q})$ through the process outlined in the proof of the Descent Theorem such that $h(R) \leq 9.1505$ and

$$P = a_1 Q_1 + a_2 Q_2 + 2^m R = a_2 Q_2 + 2^m R$$

for some $a_1, a_2, m \in \mathbb{Z}$, where the $a_1 Q_1$ term disappears because Q_1 is the identity, so $a_1 Q_1$ is also the identity. Since $e^{9.1505} \approx 9419.1488$, we can find all such possibilities for R by running

```
Points(E : Bound := 9419);.
```

This gives us many points, namely

```
(0 : 1 : 0), (0 : 1 : 1), (0 : -1 : 1), (1 : 1 : 1), (1 : -1 : 1),
(-2 : 1 : 1), (-2 : -1 : 1), (2 : 3 : 1), (2 : -3 : 1), (-3/4 : 13/8 : 1),
(-3/4 : -13/8 : 1), (4/9 : 17/27 : 1), (4/9 : -17/27 : 1), (12 : 43 : 1),
(12 : -43 : 1), (-55/49 : 603/343 : 1), (-55/49 : -603/343 : 1),
(70/121 : 811/1331 : 1), (70/121 : -811/1331 : 1), (154/25 : 2017/125 : 1),
(154/25 : -2017/125 : 1), (2193/2704 : 106079/140608 : 1),
(2193/2704 : -106079/140608 : 1), (-4472/2601 : 201709/132651 : 1),
(-4472/2601 : -201709/132651 : 1), (5304/1849 : 413867/79507 : 1),
(5304/1849 : -413867/79507 : 1).
```

That's 27 points. I can tell you for certain that, assuming that I did my calculations right with κ and κ', then Q_2 together with all the points listed above does indeed generate $E(\mathbb{Q})$, and thus it is finitely generated. Showing, however, that it is indeed generated by two elements, seems a bit more daunting. I suppose I would have to find a point that generates the subgroup $2E(\mathbb{Q})$, which I'm not even sure is cyclic in the first place. It stands to reason that if any point in $E(\mathbb{Q})$ generates all 27 of these points, then it is probably one of these points, since heights only tend to increase. I tried narrowing down which points I had to test by finding the order of each one with the `Order()` function, but that only eliminated three possibilities.

I decided to brute force it at this point and see if any of these points generated all the others using the following code, expecting a final count of 27 for a generator:

```
E := EllipticCurve([0, 1, 0, -2, 1]);
Q1 := E![0, 1, 0];
Q2 := E![-2, 1];
points := Points(E : Bound := 9419);

for point in points do;
  count := 0;
  for i in [-50..50] do;
    if i * point in points then;
      count := count + 1;
    end if;
  end for;
  printf "%o: %o\n", point, count;
end for;
```

The best this got me was a few points that generate 9 of the 27, but nothing better than that. So that being said, I'm not sure which of those points I should choose to accompany Q_2 as the second generator of $E(\mathbb{Q})$. (One of the points that generated 9 of them was $(1 : 1 : 1)$ in case that counts for something.) $\square$

2. $*$ Let $T : x^2 + y^2 = 1$ and $T(\mathbb{Q}) = \{(x, y) : x, y \in \mathbb{Q} \text{ and } x^2 + y^2 = 1\}$. (We don't include points at infinity in T.)

The set $T(\mathbb{Q})$ has the structure of a group, with the group operation defined by

$$(x_1, y_1) * (x_2, y_2) = (x_1x_2 - y_1y_2, x_1y_2 + x_2y_1).$$

(This is equivalent to adding angles, and the formula above is probably reminiscent of the addition formulas for cosine and sine.) Note: You may assume that this operation gives $T(\mathbb{Q})$ the structure of a group. You don't need to prove it.

(a) Let $S = \{p_1, \ldots, p_k\}$ be a finite set of prime numbers. Show that if Q_1 and Q_2 are points in $T(\mathbb{Q})$ where the denominator of the x and y coordinates have only prime factors from S, then $Q_1 * Q_2$ has the same property.

(b) Show that if S is any finite set of prime numbers there is some point $(x, y) \in T(\mathbb{Q})$ where the denominators of x and y contain prime factors not in S. (The parametrization of rational points on $x^2 + y^2 = 1$ might be helpful here!)

(c) Conclude that $T(\mathbb{Q})$ is not finitely generated. (The main way the descent theorem breaks in this case is because $[T(\mathbb{Q}) : 2T(\mathbb{Q})]$ is not finite.)

Proof.

(a) Write

$$Q_1 = \left(\frac{a}{p_1^{a_1} \cdots p_k^{a_k}}, \frac{b}{p_1^{b_1} \cdots p_k^{b_k}} \right), Q_2 = \left(\frac{c}{p_1^{c_1} \cdots p_k^{c_k}}, \frac{d}{p_1^{d_1} \cdots p_k^{d_k}} \right)$$

for some integers $a, b, c, d, a_1, \ldots, a_k, b_1, \ldots, b_k, c_1, \ldots, c_k, d_1, \ldots, d_k$. Then

$$\begin{aligned} &Q_1 * Q_2 \\ &= \left(\frac{ac}{p_1^{a_1c_1} \cdots p_k^{a_kc_k}} - \frac{bd}{p_1^{b_1d_1} \cdots p_k^{b_kd_k}}, \frac{ad}{p_1^{a_1d_1} \cdots p_k^{a_kd_k}} + \frac{bc}{p_1^{b_1c_1} \cdots p_k^{b_kc_k}} \right) \\ &= \left(\frac{ac \left(p_1^{b_1d_1} \cdots p_k^{b_kd_k} \right) - bd \left(p_1^{a_1c_1} \cdots p_k^{a_kc_k} \right)}{p_1^{a_1c_1+b_1d_1} \cdots p_k^{a_kc_k+b_kd_k}}, \frac{ad \left(p_1^{b_1c_1} \cdots p_k^{b_kc_k} \right) + bc \left(p_1^{a_1d_1} \cdots p_k^{a_kd_k} \right)}{p_1^{a_1d_1+b_1c_1} \cdots p_k^{a_kd_k+b_kc_k}} \right), \end{aligned}$$

which has the desired property.

(b) Let S and Q_1 have notation as in (a). Let $m = \alpha/\beta$, where $\alpha, \beta \in \mathbb{Z}$ and m is not the slope of the line tangent to $x^2 + y^2 = 1$ at Q_1. We can then find the unique point $P = (X, Y) \in T(\mathbb{Q})$ such that the line between Q_1 and P has slope m.

First, we find the y-intercept of this line, which will be

$$B = \frac{b}{p_1^{b_1} \cdots p_k^{b_k}} - \frac{a\alpha}{\beta p_1^{a_1} \cdots p_k^{a_k}} = \frac{b\beta p_1^{a_1} \cdots p_k^{a_k} - a\alpha p_1^{b_1} \cdots p_k^{b_k}}{\beta p_1^{a_1+b_1} \cdots p_k^{a_k+b_k}}.$$

Thus

$$\begin{aligned} Y &= \frac{\alpha X}{\beta} + \frac{b\beta p_1^{a_1} \cdots p_k^{a_k} - a\alpha p_1^{b_1} \cdots p_k^{b_k}}{\beta p_1^{a_1+b_1} \cdots p_k^{a_k+b_k}} \\ &= \frac{\alpha X p_1^{a_1+b_1} \cdots p_k^{a_k+b_k} + b\beta p_1^{a_1} \cdots p_k^{a_k} - a\alpha p_1^{b_1} \cdots p_k^{b_k}}{\beta p_1^{a_1+b_1} \cdots p_k^{a_k+b_k}}. \end{aligned}$$

Certainly, we can choose β such that it contains no factors in S, and then Y will be rational with denominator not consisting of prime factors only in S, as desired.

(c) Consider the subgroup G (which is a subgroup by (a)) generated by some finite set of points $P_1, \ldots, P_m \in T(\mathbb{Q})$. The set of primes represented in the denominators of $P_1, \ldots, P_m$ is a finite set (call it S), and every linear combination of $P_1, \ldots, P_m$ (i.e. every element of G) has a denominator which is a product of primes in S. By (b), then, there exist points of $T(\mathbb{Q})$ which are not in G. Since G was an arbitrary finitely generated subgroup, this shows that every finitely generated subgroup of $T(\mathbb{Q})$ is a proper subgroup, and thus $T(\mathbb{Q})$ is not finitely generated.

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